Peer Review 2

Graded

18 Hours, 17 Minutes Late

Student

Colin Skinner

Total Points

1 / 1 pts

Question 1

Peer Review 2

1 / 1 pt

✔ **+ 1 pt** Complete

+ 0.75 pts Not a substantive review

– 0 pts Incomplete

Question assigned to the following page: 1

Battling from Z to A

What is the author trying to do?
In this paper, the author constructs an agent to win battles in Pokemon Legends: Z-A, a game with a “real-time” battle system new to the Pokemon series. They frame the battle system as an MDP solved using Q-learning and Epsilon-greedy exploration, with agent performance evaluated with win-rate across thousands of randomized Pokemon and movesets.

Is the goal of this paper significant?
Previous Pokemon games have somewhat predictable outcomes because of their “turn-based” battle system, and this new installment has substantially different battle mechanics, with real-time actions, move cooldowns, and ranged attacks adding uncertainty that is perfect for an MDP. The goal of this paper is significant because it informs players on the optimal game strategy of this new battle system without the trial-and-error (and mainly, the substantial amount of playtime) needed to learn optimal strategies themselves.

Is the method of approach valid?
Given the large uncertainty of the problem, and due to the fact that the true model of the battle system is unknown (Nintendo proprietary software), Q-learning is appropriate for this context. Discretizing move-readiness into categorical bins instead of a continuous value makes computation much easier as well while still maintaining a well-modeled state space for move-readiness. I also liked the model simplification of the opponent’s move cooldown section of the state because of it being an “unknown variable.” Even though that data is known at training-time, they are hidden variables to the agent, mirroring reality.

Is the actual execution of the research correct?
From a Pokemon fan, it is well-executed and thorough in its training. It is built to handle edge cases like when both Pokemon are knocked out, and the “legality” of Pokemon having moves that it can actually learn at that level. It also models real strategies well, like how a player would aim to prioritize keeping some health rather than simply knock their Pokemon out.

Are the correct conclusions being drawn from the results?
The conclusions reasonably follow the results, with performance of individual Pokemon discussed as well as the impact of the randomness of dodging. It is understandable that maximizing type matchups remains the most useful strategy, as it is in previous Pokemon installments, and the results heavily support this conclusion (e.g fire type moves are strong against four types, while electric moves are strong against two, so it wins more often). It would be interesting to analyze which strategies caused the agent to lose a battle, like the performance of using a specific move, which means that this simulator could even model the general effectiveness of each individual move, as learned by random assortments of Pokemon. Given the

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incredible complexity of modeling the new battle system, I am impressed by how well the agent performs, with win rates all above 68%, with one reaching as high as 78%.

Is the presentation satisfactory?
I am a Pokemon fan, but have never played Legends: Z-A, and this paper is a good accessible introduction to this battle system that is very easy-to-read. The discussion of future work by the author is strong, with extension to teams of Pokemon instead of single battles and other state considerations like the opponent Pokemon's type and species. This shows good awareness of the project's limitations and ways in which the "simplifications" of the model could be improved upon.